

Nilesh Raj

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SUMMARY

AI/ML Engineer & Forward Deployed Engineer delivering production-grade agentic AI applications directly to client environments. Architected multi-agent workflows (LangGraph, CrewAI), RAG pipelines, and traditional ML models (BERT, PyTorch), wrapping them into secure, scalable microservices using FastAPI and Docker. Developed interactive Streamlit user interfaces to validate model performance, drive stakeholder alignment, and streamline enterprise infrastructure adoption.

EDUCATION

United University Prayagraj, UP

Jun 2022 – May 2025

BCA with IBM Specialization (AI & DS)

PROJECTS

OMNICREW AI — Autonomous Web Extraction Platform | *Python, CrewAI, FastAPI, Docker* 2026

- Orchestrated a multi-agent RPA platform using CrewAI to autonomously extract and structure unstructured web data in real time, cutting data-collection time well below manual scraping.
- Engineered a resilient LLM routing pipeline (Mistral to OpenAI) with Redis-backed caching, eliminating API calls for repeated queries.
- Deployed the platform via Docker and FastAPI, using WebSockets for real-time agent telemetry streaming to a React/TypeScript frontend, giving stakeholders live visibility into pipeline operations.
- Implemented semantic deduplication and AWS S3 archival for enterprise-scale data ingestion, minimizing redundant storage before archival.
- [Deployed App](#) | [Code](#)

LumanGuide — Agentic RAG System | *Python, LangGraph, FastAPI, FAISS, Streamlit, Claude Code* 2026

- Architected an enterprise-grade Agentic RAG system on a LangGraph state machine, orchestrating intelligent query routing and dynamic document grading to improve retrieval speed and reduce latency.
- Hardened the application for client integration with AppSec measures including JWT RBAC, prompt injection guardrails, and automated secret scanning during multimodal PDF ingestion.
- Integrated LangSmith for LLM Ops tracing and connected a Streamlit dashboard via REST API, enabling non-technical stakeholders to monitor retrieval accuracy and token usage.
- [Live App](#) | [Code](#)

EXPERIENCE

AI/ML Trainee (Internship)

Feb 2026 – May 2026

Technosavvys Education Technology

Remote

- Developed a text classifier using BERT and Hugging Face to automate sentiment analysis of 500+ student course reviews, achieving 88% accuracy through iterative hyperparameter tuning.
- Streamlined instructional workflows by deploying a real-time Streamlit dashboard backed by SQLite, reducing manual review time for 10+ mentors across 5+ active courses.
- Optimized NLP preprocessing pipelines, meaningfully improving baseline model accuracy and accelerating the generation of weekly feedback reports.

TECHNICAL SKILLS

AI & GenAI: LangGraph, CrewAI, LangChain, Agentic AI Systems, RAG Pipelines, BERT, Transformers, Semantic Chunking, Embedding Models, Vector Search, NVIDIA NIM, LangSmith

Deployment & Infrastructure: Docker, Kubernetes, FastAPI, Streamlit, REST APIs, WebSockets, Redis, AWS (S3, EC2, Lambda), Git, CI/CD, n8n Automation, Anthropic Claude API

Data & ML Engineering: Python, SQL, PyTorch, Scikit-learn, Hugging Face, Pandas, NumPy, Pydantic, Vector Databases, PostgreSQL, MongoDB, SQLite